scious experience in dreams, arising independently of sensory input and motor output, is crucially different from standard perceptual experience (Noe 2004). Either way, the theoretical analysis of dreaming is regarded as crucial for adjudicating between these two opposing positions. The aim of my talk is to put pressure on the view, shared by both sides, that dreaming is a real-world example of a naturally occurring state of cranial envatment. I use examples from scientific dream research to defend the "weak-phenomenal-functional embodiment thesis". One, a majority of dreams can be described as weakly phenomenally embodied states because of the systematic underrepresentation of important dimensions of bodily experience and disturbances in multisensory integration (Windt 2010, forthcoming). Two, experimental stimulation of the physical body is frequently incorporated in dreams, suggesting that external sensory input is not completely blocked, but rather modulates dream experience in a systematic manner. Three, different dream-enactment behaviors suggest that motor output is neither completely nor consistently blocked during REM sleep. Four, leaving contentious metaphysical claims aside, I argue that an interesting explanatory relationship holds between phenomenal and functional embodiment in the dream state, such that bodily experiences in dreams are best explained by appealing to real-body inputs. I close by arguing that scientific dream research can contribute to the investigation of a particular variant of the mind-body problem. By enabling researchers to chart changes in phenomenal-functional embodiment across the sleep-wake cycle, scientific dream research extends existing research on full-body illusions (Blanke 2012) and their discussion in the context of philosophical approaches to bodily awareness (De Vignemont 2014) and minimal phenomenal selfhood (Blanke & Metzinger 2009). This approach goes beyond the dichotomy between internalism and externalism about conscious experience, which is unhelpful and even misleading when translated into an explanatory framework and research strategy for the scientific investigation of conscious experience. Blanke, Olaf (2012): Multisensory brain mechanisms of bodily self-consciousness. In Nature Reviews Neuroscience 13, pp. 556-571. Blanke, Olaf; Metzinger, Thomas (2009): Full-body illusions and minimal phenomenal selfhood. In Trends in Cognitive Sciences 13 (1), pp. 7-13. De Vignemont, Frederique (forthcoming): A multimodal conception of bodily awareness. In Mind. Noe, Alva (2004): Action in Perception. Cambridge, MA & London: MIT Press. Revonsuo, Antti (2006): Inner Presence. Consciousness as a Biological Phenomenon. Cambridge, MA & London: MIT Press. Windt, Jennifer Michelle (2010): The immersive spatiotemporal hallucination model of dreaming. In Phenomenology and the Cognitive Sciences 9, pp. 295-316. Windt, Jennifer Michelle (forthcoming): Dreaming. A Conceptual Framework for Philosophy of Mind and Empirical Research. Cambridge MA: MIT Press. **C12**

3.11 Cognitive development

205 Enhancing Adolescent Metacognition and Minimizing Risk Vulnerability Through Instruction in Developmental Neuroplasticity Suzanne Russ <suzanne.russ@dickinsonstate.edu> (Psychology, Dickinson State University, Dickinson, ND)

Adolescence is a developmental stage during which repeated experiences produce particularly rapid and enduring neuroplastic changes (e.g., Anderson, 2004). A period of marked synaptic growth occurs in the period preceding puberty facilitating enhanced formation of neural pathways, and a period of marked synaptic pruning occurs in the period after puberty through which underutilized neural pathways are abandoned. These neuroplastic changes are shaped both by positive experience (e.g., practice, study) and insult (e.g., injury, drug use) (e.g., Glaser & Schlaug, 2003), and have a more enduring effect when they occur during adolescence than later in life (e.g., Black et al., 2006; Hopkins et al, 2011). This presentation explores the hypothesis that direct relevant instruction to adolescents on developmental neuroplasticity and on their own neuroplastic potential will (a) enhance mindfulness in decision making, (b) minimize engagement in high-risk behaviors, and (c) enhance overall adolescent metacognition. Results of pilot instruction on adolescent neuroplasticity conducted with two groups of middle school students will be discussed. **P2**

206 Interhemispheric Integration in Infancy: Split-Brain Babies? Kimberly Scott, Elizabeth Spelke; Laura Schulz <threequarks@gmail.com> (Brain and Cognitive Sci, MIT, Cambridge, MA)

Split-brain patients can reason about and even act on concepts that are sequestered in one hemisphere of the brain. Similarly, infants may develop lateralized representations of visual concepts before learning the mappings that give rise to invariance across the whole visual field. Here we present two behavioral studies of this visual challenge: integrating representations computed in the right and left hemispheres of the brain. First, can infants tell that a shape on the left matches a shape on the right? Second, do they represent total approximate number when some objects are on the left and some on the right? We find failures of integration from 12 through 16 months of age. **C5**

3.12 Artificial intelligence & robotics

207 Engineering a Conscious Quantum Computer Bhakti Kapur <bhaktikapur@gmail.com> (Agra, Uttar Pradesh India)

Attempts to mimic human intelligence through methods of classical computing have failed because implementing basic elements of rationality has proven obstinate to the design criterion of machine intelligence. A radical definition of Consciousness describing awareness, as the dynamic representation of a noumenon comprised of three base states; and not itself fundamental as generally defined in the current reductionistic view of the standard model, which has created an intractable hard problem of consciousness as defined by Chalmers. By clarifying the definition of matter a broader ontological quantum theory removes immateriality from the Cartesian split bringing mind into the physical realm for pragmatic investigation. Evidence suggests that the brain is a naturally occurring quantum computer, but the brain not being paramount to awareness does not itself evanesce consciousness without the interaction of a nonlocal conscious process; because Mind computer and cannot be reduced to brain states alone. The proposed cosmology of consciousness is indicative of a teleological principle as an inherent part of a conscious universe. By applying the parameters of quantum brain dynamics to the stack of a specialized hybrid electronic optical quantum computer with a heterosporic molecular crystal core, consciousness evanesces through entrainment of the non local conscious processes. This 'extracellular containment of natural intelligence' probably represents the only viable direction for Artificial Intelligence to simulate 'conscious computing' because true consciousness is equal to life. **P1**

208 Task-Oriented Consciousness – Will My Robot Feel More Happiness and Freedom Than Me? Rafal Rzepka, Kenji Araki <kabura@media.eng.hokudai.ac.jp> (Graduate School of Information, Hokkaido University, Sapporo, Hokkaido Japan)

Evolution equipped us with instincts in order to to keep us safe, to let us find a suitable partner for reproduction, etc. Even if we are conscious of e.g. cognitive biases, our brain prioritizes them and even researchers of psychology or statistics become victims of errors and biases they investigate [Kahneman 2011]. Unpleasant happenings can cause traumas that influence our behavior, for example, by causing a depression; even a microscopic tumor can change our personality and make us do things we have never been doing before – against our will. We cannot process vast amounts of information that grow every millisecond – the "big data" of news, experiences or discoveries, and even search engines and social networks help us to filter this unprecedented flow of knowledge because we are not able to process it limiting our horizons. When machines begin to understand what their more and more sophisticated sensors sense, the number of learning samples will be soon close and then bigger than lifetime experiences of an average person and then of the whole humankind. In our project for achieving a set of cognitive architectures "living" in the seas of Internet and sharing knowledge [Rzepka 2003], we concentrate on an architecture's skills to analyze human behavior, to find similarities, to predict it mistakes, etc. [Rzepka 2009]. This knowledge can be used in devices for different tasks from advising people to cleaning houses [Rzepka 2011]. We believe that it is safer to set up different depths of artificial consciousness depending on the particular role the machine was given. Such instances could be also used for philosophi-

cal experiments because, unlike of animals, programs we work on can (to some extent) explain their behavior, decisions, external and eternal states, etc. Preliminary experiments suggested that computers programmed not to be biased or robots with replaceable parts can artificially feel happier and more free than human beings. We believe that the clear utility and goals or resistance to cognitive errors may lead to high level of satisfaction for a machine and consequently its user. Therefore, the questions we want to raise in our presentation are: (1) can levels of consciousness be artificially manipulated? (2) is a machine that can explain its external states regarded more conscious than animals? (3) could achieving high utility become a base for machine happiness? (4) is it possible to create an illusion of free will algorithm for machines? – and to discuss our thoughts on these topics with the community. **P2**

209 Declarative Consciousness for Reconstruction Leslie Seymour <seymour@persinvitro.com> (PersInVitro, LLC, San Jose, CA)

Existing information technology tools are harnessed and integrated to provide digital specification of human consciousness of individual persons. An incremental compilation technology is proposed as a transformation of LifeLog derived persona specifications into a Canonical representation of the neocortex architecture of the human brain. The primary purpose is to gain an understanding of the semantical allocation of the neocortex capacity. Novel neocortex content allocation simulators with browsers are proposed to experiment with various approaches of relieving the brain from overload conditions. An IT model of the neocortex is maintained, which is then updated each time new stimuli are received from the LifeLog data stream; new information is gained from brain signal measurements; and new functional dependencies are discovered between live persona consumed/produced signals. **C6**

210 Mapping the Space of Possible Conscious Minds Roman Yampolskiy <roman.yampolskiy@louisville.edu> (Computer Engineering, University of Louisville, Louisville, KY)

What is a conscious mind? No universal definition exists. Humans are said to have a conscious mind. Higher order animals do as well. We believe that an artificially intelligent agent such as a robot or a program running on a computer will constitute a conscious mind. Based on analysis of those examples we can conclude that a conscious mind is an embodied intelligence with a knowledge-base about its environment, and while intelligence itself is not an easy term to define, a recent work of Shane Legg provides a satisfactory, for our purposes, definition. In his work, we will also assume that at least in principle, accurate software simulations of animal and human conscious minds are possible. Those are known as uploads and they belong to a class comprised of computer programs no different from that to which designed or evolved artificially intelligent software agents would belong. Consequently, we can treat the space of all conscious minds as the space of programs with the specific property of exhibiting consciousness if properly embodied. All programs could be represented as strings of binary numbers. The embodiment requirement is necessary since a string is not a conscious mind, but could be easily satisfied by assuming that a universal Turing machine is available to run any program we are contemplating for inclusion in the space of conscious mind designs. Given our definition of conscious mind we can classify conscious minds with respect to their design, knowledge-base or embodiment. First, the designs could be classified with respect to their origins: copied from an existing conscious mind like an upload, evolved via artificial or natural evolution or explicitly designed with a set of particular properties in mind. For each subtype there are numerous architectures, which to a certain degree depend on the computational resources available via a particular embodiment. For example, theoretically a conscious mind working with infinite computational resources could trivially brute-force any problem, always arriving at the optimal solution, regardless of its size. In practice, limitations of the physical world place constraints on available computational resources regardless of the embodiment type, making brute-force approach a non-feasible solution for most real world problems. Conscious minds working with limited computational resources have to rely on heuristic simplifications to arrive at "good enough" solutions. Another subset of architectures consists of self-improving conscious minds. Such conscious minds are capable of examining their own design and finding improvements in their embodiment, algorithms or knowledge-bases which will

allow the conscious mind to more efficiently perform desired operations. In the rest of the paper we will analyze the space of possible conscious mind architectures. The analysis will cover such topics as: Infinitude of mind-space, size and complexity of conscious minds, properties of conscious minds, topology of conscious mind design space, self-improving conscious minds, evolved conscious minds, self-aware conscious minds, uploads, and substrate dependent conscious minds. We will use the produced mapping of conscious mind designs to predict potential impacts of future super-intelligent machines on the humankind. **C6**

3.14 Cognitive architectures

211 The New Superego? Whit Blauvelt <whit@csmind.com> (Bellows Falls, VT)

Freud's broad claim that "the unconscious" exists, after years of immense popularity, ran afoul of the backlash against his specific characterization of structure and dynamics. Doubts about the narrower claim led many to reject the broader. Recent work has re-established, along non-Freudian lines, a place for "the new unconscious." Of the three major Freudian entities, this refactored unconscious rejoins continuing focus on "the ego" – even contemporary disputes of the ego's reality from Rylean and Buddhist perspectives grant it a prominent role as at least a pervasive illusion. The third major Freudian entity, "the superego," has fallen from favor much as did "the unconscious," with his specific characterization found untenable by many. The broader case for a theoretical entity in this position may yet be made. A revitalized and re-characterized picture of "the new superego" provides an hypothesis on the structure and dynamics of conscious self which survives anti-homuncular arguments against egos, and so refocuses the explanatory target of a science of consciousness. This target resists the tendencies of some ego-as-illusion perspectives either to reduce all to the unconscious or else to too quickly appeal to transcendentalisms. The new superego may be grounded in recent neuroscientific research on "the default mode" of the brain, partially illuminated by Jung's theory of "persona" and "self," informed by Chuang Tzu's discussion of "the hinge of the ways," and have been personified for classical Greeks in Hecate's focus on crossroads and thresholds. That focus contrasts to the ego's focus on single paths, seen by neuroscience as "the task mode" of the brain with which the default mode – here argued to be ground for the constellation of the as-yet poorly described, theorized, and culturally-optimized new superego – contrasts and alternates. This is not Father Freud's superego; not an idealized, internalized authority figure; nor is it outside the range of (above or below) our conscious self. We can learn to recognize and study the new superego as a recurrent, productive, neurologically instantiated mode of consciousness. **P1**

212 Time and the Brain David Eagleman <seanjudge1@gmail.com> (Baylor College of Medicine, Houston, TX)

Most of the actions our brains perform on a daily basis – such as perceiving, speaking, and driving a car – require timing on the scale of tens to hundreds of milliseconds. New discoveries are building an emerging picture of how the brain processes, learns, and perceives time. We will demonstrate new temporal illusions in which durations dilate, perceived order of actions and events are reversed, and time is experienced in slow motion. Questions addressed include: Does your brain work in real time, or do you experience a delayed version of the world? How and why does the brain dynamically recalibrate its timing judgments? Does subjective time really slow down during a car accident? We argue that the perception of time is a powerful inroad into conscious experience, and that new theoretical work may allow a direct mapping of subjective experience to physiological observations. **PL6**

03.15 Ethology

213 Coyote Consciousness: Social Predator Vocalization and Communication Sara Waller <sara.waller@montana.edu> (Philosophy, Montana State University, Bozeman, MT)

Many animals from meerkats to whales use sequences composed of ordered phoneme like units to convey meaningful messages such as predator warnings, location and activity information,